A Biologically-Inspired Framework for Online Mining Spatio-Temporal Dependency Rules from Event Data Streams

POSTER

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An event-driven neural network maintains spatial dependency rules as events arrive. Connection weights record dependency strength and decay over time.

Problem and rule representation

Each event has a type, a location and a timestamp. We look for event types that tend to follow one another in particular places, and update these dependencies as the stream changes.

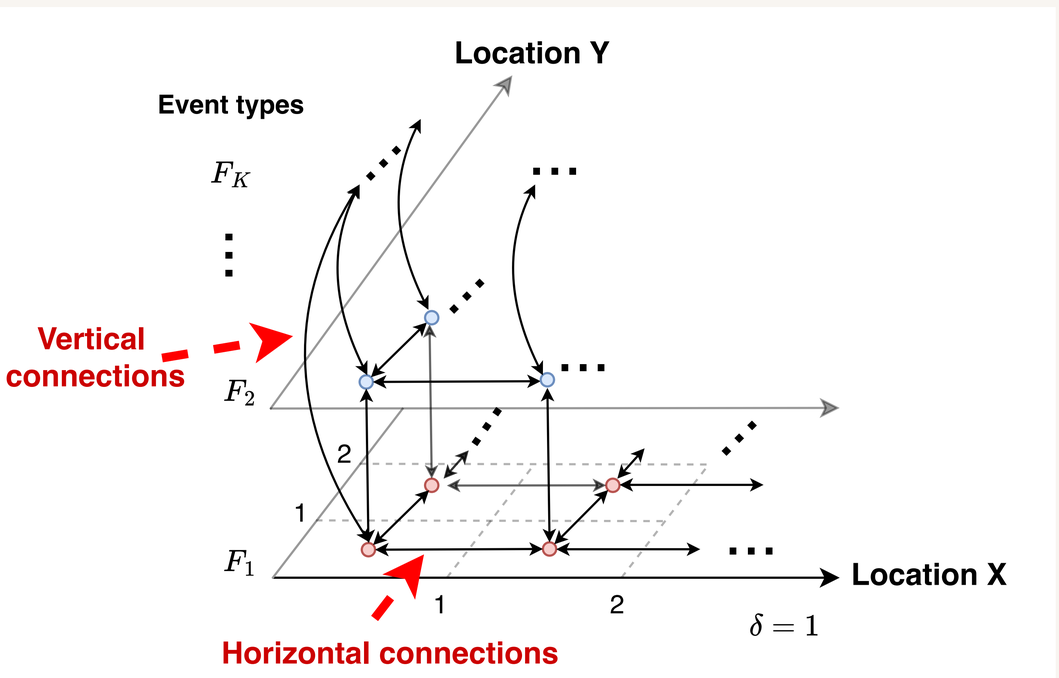
A **spatio-temporal dependency rule (STDR)** links two event-type/location pairs:

$$(F_i, \mathrm{loc}_i) \xrightarrow{w_{ij}} (F_j, \mathrm{loc}_j).$$

The weight w_{ij} gives the rule's strength. For example, office events in one cell may precede subway events in another. These rules describe statistical dependencies, not causal effects.

Network architecture

We divide the region into a rectangular grid and place one neuron for each event type in each cell. An incoming event makes its corresponding neuron fire.



Neurons connect within each cell and across cells whose centers lie within distance δ. Each pair has two directed connections, whose weights can differ.

Learning and rule selection

We initialize weights to zero and update them with an unsupervised, Hebbian-inspired learning rule:

$$\frac{dw_{ij}}{dt} = \underbrace{\eta A}_{\text{reinforcement}} - \underbrace{\eta R w_{ij}}_{\text{scaling}} - \underbrace{\lambda w_{ij}}_{\text{forgetting}} .$$

A and R record recent presynaptic and postsynaptic activity. Both traces decay exponentially, with timescales τ_A and τ_R. The learning rate is η, and $\lambda > 0$ controls long-term decay.

For the continuous learning equation, $0 \le A \le 1$ and $R \ge 0$ give $w_{ij} \le w_{\max} = \eta/\lambda$. We report a rule as significant when

$$w_{ij} \ge \theta w_{\max}, \qquad 0 < \theta < 1.$$

Online updates

1. Map the incoming event to its neuron and emit a spike.
2. Select the firing neuron's incoming and outgoing connections, plus currently significant connections.
3. Decay the selected weights and traces using the elapsed time, then apply the learning update.
4. Remove connections below the significance threshold and report the remaining rules.

Time per event $O(d_{\max} + |\mathbf{W}|)$
Memory $O(|\mathbf{E}| + |\mathbf{W}|)$

$d_{\max}$ is the maximum total in- and out-degree, $\mathbf{E}$ is the set of all connections, and $\mathbf{W}$ is the set of currently significant connections.

Evaluation

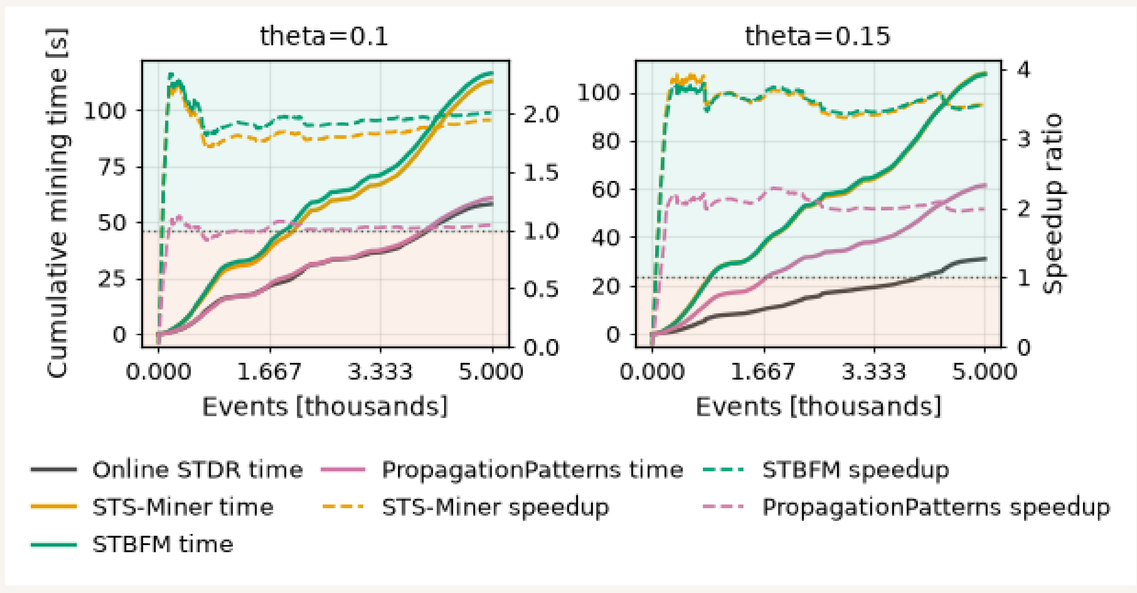
227,428
NYC Foursquare check-ins

251
event types

We compare with **STS-Miner, STBFM and Propagation Patterns**:

- Rerun the baselines on a 120-minute moving window.
- Restrict patterns to pairs or two-node structures.
- Remove spatial annotations, merge equivalent STDRs and tune baseline thresholds to return roughly similar numbers of patterns.

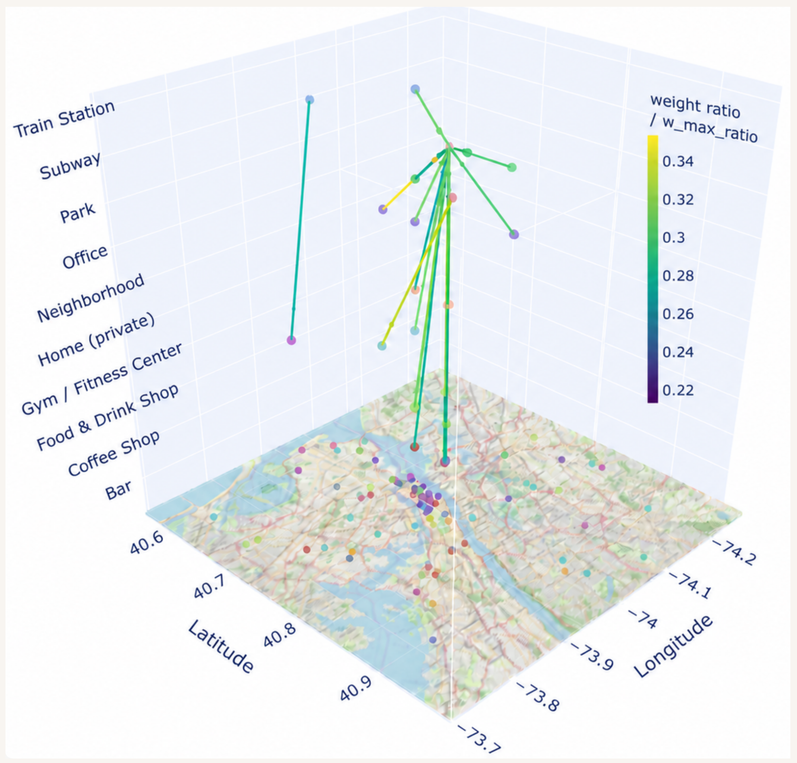
Mining time



Cumulative mining time (solid) and baseline-to-online speedup (dashed) for $\theta = 0.10$ and 0.15. Ratios above one favor the online miner.

The online miner is faster in most tested settings, especially at higher significance thresholds. This compares runtime across methods with different pattern semantics.

Example spatial rules



Snapshot for 18:00–21:00 with $\theta = 0.2$. Locations form the base plane, and event types appear on the vertical axis.

The strongest reported rule links an **Office cell in Brooklyn** with a **Subway cell in Manhattan**. The prototype lets users select event types and filter the strongest rules.

Conclusions and limitations

The network maintains spatially explicit rules as events arrive. Runtime was lower in most tested settings. The current method finds **pairwise dependencies**. Results depend on the grid, neighborhood radius and learning parameters. Evaluation covers one application domain, and rule weights can be difficult to interpret.

References Huang, Zhang and Zhang (2008), *IEEE TKDE* 20(4), STS-Miner. Maciąg, Kryszkiewicz and Bembenik (2019), *KES*, closed spatio-temporal sequential patterns. Moosavi et al. (2019), *KDD*, short- and long-term pattern discovery. Yang et al. (2014), *IEEE TSMC: Systems*, LBSN activity data.

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Code and supplementary material

github.com/piotr-maciag/Online_STDR_Miner